

3	(a)	By Newton's Third Law, when both bodies collide, the two forces exerted by both bodies on each other are equal in magnitude and opposite in direction, hence $F_{21} = -F_{12}$ [M1] By Newton's Second Law, the (resultant) force on a body is equal to the rate of change of momentum, i.e. $F = \frac{\Delta p}{\Delta t}$, hence $\frac{\Delta p_1}{\Delta t} = -\frac{\Delta p_2}{\Delta t}$ [M1] where Δp_1 is the change in momentum for m_1, Δp_2 is the change in momentum for m_2 Since the time of collision Δt is the same for bodies, the change in momentum for both bodies are equal and opposite, hence $\Delta p_1 = -\Delta p_2$ $m_1(v_1 - u_1) = -m_2(v_2 - u_2)$ $m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$ Hence the total vector sum of the change in momentum of the system is zero. [A1] And since F_{12} and F_{21} are internal forces for the system, there is no external resultant force on the system.	
3	(b)	(i) Identical blocks moving at same speed so meet at mid-point. Distance travelled by a block = 0.450 0.025 = 0.425 m [M1] $\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{0.425}{0.18} = 2.4 \text{ s} \quad [\text{A1}]$ or Distance between surfaces of the blocks = 0.900 – 0.050 = 0.850 m Relative speed between blocks = $v_A - v_B = 0.18 - (-0.18) = 0.36 \text{ m s}^{-1}$ [M1] $\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{0.850}{0.36} = 2.4 \text{ s} \quad [\text{A1}]$	
3	(b)	(ii) $KE_{\text{initial}} = \frac{1}{2}(0.17)(0.18^2) = 0.002754 \text{ J}$ $KE_{\text{final}} = \frac{1}{2}mv_f^2 = \frac{80}{100}(0.002754) \quad [\text{M1}]$ $v_f = \sqrt{\frac{\frac{80}{100}(0.002754)}{\frac{1}{2}(0.17)}}$ $= 0.16 \text{ m s}^{-1} \quad [\text{A1}]$	

3	(b)	(iii)	$\Delta v = v_f - v_i = 0.16 - (-0.18) = 0.34 \text{ m s}^{-1} \quad [\text{M1}]$ Impulse = $F_{\text{ave}} \Delta t = m \Delta v$ $F_{\text{ave}} = \frac{m \Delta v}{\Delta t}$ $= \frac{0.17(0.34)}{0.07} \quad [\text{M1}]$ $= 0.83 \text{ N}$ $F_{\text{max}} = 2F_{\text{ave}} = 1.7 \text{ N} \quad [\text{A1}]$ Or area under graph = impulse = change in momentum $\frac{1}{2}(F_{\text{max}})(\Delta t) = m \Delta v$ $F_{\text{max}} = \frac{2m \Delta v}{\Delta t}$ $= \frac{2(0.17)(0.34)}{0.07} \quad [\text{M1}]$ $= 1.7 \text{ N} \quad [\text{A1}]$	
4	(a)	(i)	energy / kJ	